

TASK 1

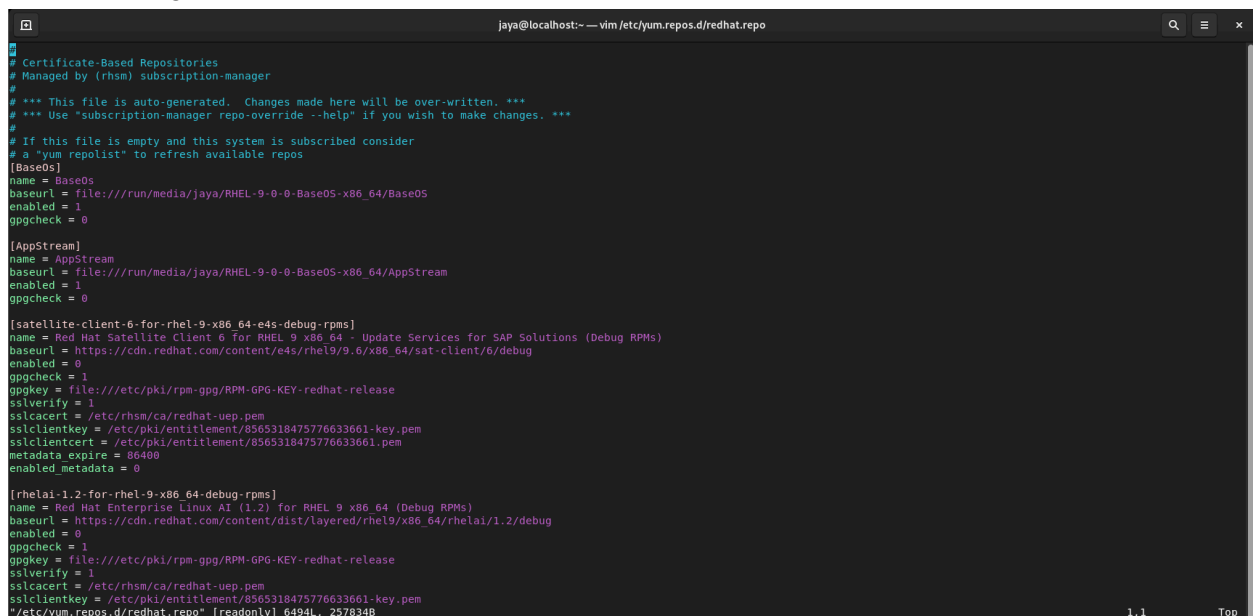
Task 1.1 Creating reposerver using ISO file

Objective: The goal is to set up Repo server in a network so we can use

prerequisites

- VMware
- VMware Vnet
- RHEL VM
- RHEL iso file

1. Configure server to use repos from ISO files



```
jaya@localhost:~ — vim /etc/yum.repos.d/redhat.repo
# Certificate-Based Repositories
# Managed by (rhsm) subscription-manager
#
# *** This file is auto-generated.  Changes made here will be over-written. ***
# *** Use "subscription-manager repo-override --help" if you wish to make changes. ***
#
# If this file is empty and this system is subscribed consider
# a "yum repolist" to refresh available repos
[BaseOS]
name = BaseOS
baseurl = file:///run/media/jaya/RHEL-9-0-0-BaseOS-x86_64/BaseOS
enabled = 1
gpgcheck = 0

[AppStream]
name = AppStream
baseurl = file:///run/media/jaya/RHEL-9-0-0-BaseOS-x86_64/AppStream
enabled = 1
gpgcheck = 0

[satellite-client-6-for-rhel-9-x86_64-e4s-debug-rpms]
name = Red Hat Satellite client 6 For RHEL 9 x86_64 - Update Services for SAP Solutions (Debug RPMs)
baseurl = https://cdn.redhat.com/content/e4s/rhel9/9.6/x86_64/sat-client/6/debug
enabled = 0
gpgcheck = 1
gpgkey = file:///etc/pki/rpm-gpg/RPM-GPG-KEY-redhat-release
sslsverify = 1
sslcacert = /etc/rhsm/ca/redhat-uep.pem
sslclientkey = /etc/pki/entitlement/8565318475776633661-key.pem
sslclientcert = /etc/pki/entitlement/8565318475776633661.pem
metadata_expire = 86400
enabled_metadata = 0

[rhelai-1.2-for-rhel-9-x86_64-debug-rpms]
name = Red Hat Enterprise Linux AI (1.2) for RHEL 9 x86_64 (Debug RPMs)
baseurl = https://cdn.redhat.com/content/dist/layered/rhel9/x86_64/rhelai/1.2/debug
enabled = 0
gpgcheck = 1
gpgkey = file:///etc/pki/rpm-gpg/RPM-GPG-KEY-redhat-release
sslsverify = 1
sslcacert = /etc/rhsm/ca/redhat-uep.pem
sslclientkey = /etc/pki/entitlement/8565318475776633661-key.pem
"/etc/yum.repos.d/redhat.repo" [readonly] 6494L, 2578348
1,1 Top
```

2. Copy the repos which you want to host in network to /var/www/html/server

- 2.1 mkdir /var/www/html/server
- 2.2 lsblk
- 2.3 copy mount location of sr0
- 2.4 cd /run/media/jaya and ls REDHAT
- 2.5 cp BaseOS and AppStream to /var/www/html/server

```
Activities Terminal Sep 23 10:18
jaya@localhost: /var/www/html/server
[jaya@localhost server]$ cd /var/www/html/server/
[jaya@localhost server]$ ls
AppStream BaseOS
[jaya@localhost server]$
```

3. On Clint side configure

In /etc/yum.repos

Create repo

[BaseOs]

name= BaseOs

Baseurl = <http://192.168.233.130/server/BaseOS>

Enabled = 1

Gpgcheck = 0

[AppStream]

name= AppStream

Baseurl = <http://192.168.233.130/server/AppStream>

Enabled = 1

Gpgcheck = 0

```
[BaseOS]
name = BaseOs
baseurl = http://192.168.233.130/server/BaseOS
enabled = 1
gpgcheck = 0

[AppStream]
name = AppStream
baseurl = http://192.168.233.130/server/AppStream
enabled = 1
gpgcheck = 0
```

(Disable enabled repos before adding new repos)

Restart Httpd and try installing something using dnf

```

Updating Subscription Management repositories.
Last metadata expiration check: 16:03:46 ago on Mon 22 Sep 2025 06:22:21 PM IST.
Dependencies resolved.
=====
Package                        Architecture      Version           Repository        Size
=====
Installing:
httpd                          x86_64            2.4.51-7.el9_0    AppStream         1.5 M
Installing dependencies:
apr                            x86_64            1.7.0-11.el9      AppStream         127 k
apr-util                       x86_64            1.6.1-20.el9      AppStream         98 k
apr-util-bdb                   x86_64            1.6.1-20.el9      AppStream         15 k
httpd filesystem               noarch            2.4.51-7.el9_0    AppStream         17 k
httpd-tools                    x86_64            2.4.51-7.el9_0    AppStream         88 k
mailcap                         noarch            2.1.49-5.el9      BaseOS            35 k
redhat-logos-httpd            noarch            90.4-1.el9        AppStream         18 k
Installing weak dependencies:
apr-util-openssl               x86_64            1.6.1-20.el9      AppStream         17 k
mod_http2                      x86_64            1.15.19-2.el9     AppStream         153 k
mod_lua                        x86_64            2.4.51-7.el9_0    AppStream         63 k

Transaction Summary
=====
Install 11 Packages

Total download size: 2.1 M
Installed size: 6.0 M
Is this ok [y/N]: y
Downloading Packages:
[MIRROR] mailcap-2.1.49-5.el9.noarch.rpm: Status code: 404 for http://192.168.233.130/server/BaseOS/Packages/mailcap-2.1.49-5.el9.noarch.rpm (IP: 192.168.233.130)
[MIRROR] apr-1.7.0-11.el9.x86_64.rpm: Status code: 404 for http://192.168.233.130/server/AppStream/Packages/apr-1.7.0-11.el9.x86_64.rpm (IP: 192.168.233.130)
[MIRROR] apr-util-1.6.1-20.el9.x86_64.rpm: Status code: 404 for http://192.168.233.130/server/AppStream/Packages/apr-util-1.6.1-20.el9.x86_64.rpm (IP: 192.168.233.130)
[MIRROR] mailcap-2.1.49-5.el9.noarch.rpm: Status code: 404 for http://192.168.233.130/server/BaseOS/Packages/mailcap-2.1.49-5.el9.noarch.rpm (IP: 192.168.233.130)
[MIRROR] apr-1.7.0-11.el9.x86_64.rpm: Status code: 404 for http://192.168.233.130/server/AppStream/Packages/apr-1.7.0-11.el9.x86_64.rpm (IP: 192.168.233.130)
[MIRROR] apr-util-1.6.1-20.el9.x86_64.rpm: Status code: 404 for http://192.168.233.130/server/AppStream/Packages/apr-util-1.6.1-20.el9.x86_64.rpm (IP: 192.168.233.130)
[FAILED] apr-1.7.0-11.el9.x86_64.rpm: No more mirrors to try - All mirrors were already tried without success
(2-3/11): apr-util-1.6.1-20.el9.x86_64.rpm 5% [==
The downloaded packages were saved in cache until the next successful transaction.
You can remove cached packages by executing 'dnf clean packages'.
Error: Error downloading packages:
  apr-1.7.0-11.el9.x86_64: Cannot download, all mirrors were already tried without success
[root@localhost ~]# _

```

Task 1.2 Creating Reposever using redhat CDN

prerequisites

- VMware
- VMware Vnet
- RHEL VM
- Redhat subscription on repos-server

1. Download the required repo by using reposync and you can get repo id from /yum.repo.d/redhat.repo

```

Sep 24 16:33
root@localhost:var/www/html
[root@localhost html]# yum reposync -p server --repo=satellite-utils-6.17-for-rhel-9-x86_64-debug-rpms --download-metadata

```

2. Once downloaded it in server folder client can access the repo after the configuration is completed in client machine

```
[satellite]
name=satellite
baseurl=http://192.168.233.130/server/satellite/
enabled=1
gpgcheck=0
~
~
~
~
~
~
~
```

Task 2 (Additional) Automating client configuration

The script below retrieves the list of repos inside reposerver (/server) and creates repo_name.repo folder inside /etc/yum.repos.d/

```
#!/bin/bash
SERVER_IP="192.168.233.130"
REPO_DIR_URL="http://$SERVER_IP/server"
repos=$(curl -s $REPO_DIR_URL/ | grep -oP '(?<=href=)"[^"]+/' | grep -vE '^(\.|\./)$' | sed 's:/:::')
for repo in $repos; do
    cat <<EOF | sudo tee /etc/yum.repos.d/$repo.repo
[$repo]
name=$repo
baseurl=$REPO_DIR_URL/$repo/
enabled=1
gpgcheck=0
EOF
done
~
~
```

This script will be present in server and client will download it and run and automatically all the repos inside server will be configured in client machine and same script can be used to update packages

